# Certified Expression Intervals under Shared Segmentation Uncertainty

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**Abstract**

A shared cell-boundary error can alter several expression counts together in imaging-based spatial transcriptomics. We introduce GeoCert, an exact method for determining which observed cell-by-gene counts remain invariant under a physical boundary-deformation budget and which can change. For ordered boundaries subject to slope, band, thickness and nuclear constraints, one polynomial-size rational cut represents the minimum continuous deformation of every feasible molecular assignment. In the single-interface point-transcript setting, an exact critical-height construction yields a polynomial-time count dynamic program that recovers outcomes missed by every penalty bound. Disjoint boundary ribbons combine into one topology-preserving tissue partition, and min-plus composition allocates a shared area budget across them. Applied to Xenium lung crops and a breast format-test dataset, GeoCert constructs 544 and 1,708 ribbons and computes 210,240 exact interval rows. Under the primary atlas, which allows up to 0.45 µm of boundary movement along about two-thirds of the boundary length around eligible cells, 84.1% of lung and 77.1% of breast cell–gene counts are invariant at a 4 µm² budget; a wider-ribbon study quantifies the trade-off between reach and coverage. Among questions with at least two individually movable transcripts, shared geometry tightens independent bounds for 4 of 46 lung and 203 of 1,026 breast questions. Attaining boundaries certify each interval endpoint and reveal when several molecular transfers share one deformation. GeoCert turns a specified segmentation uncertainty model into exact expression-count intervals and inspectable geometric witnesses.

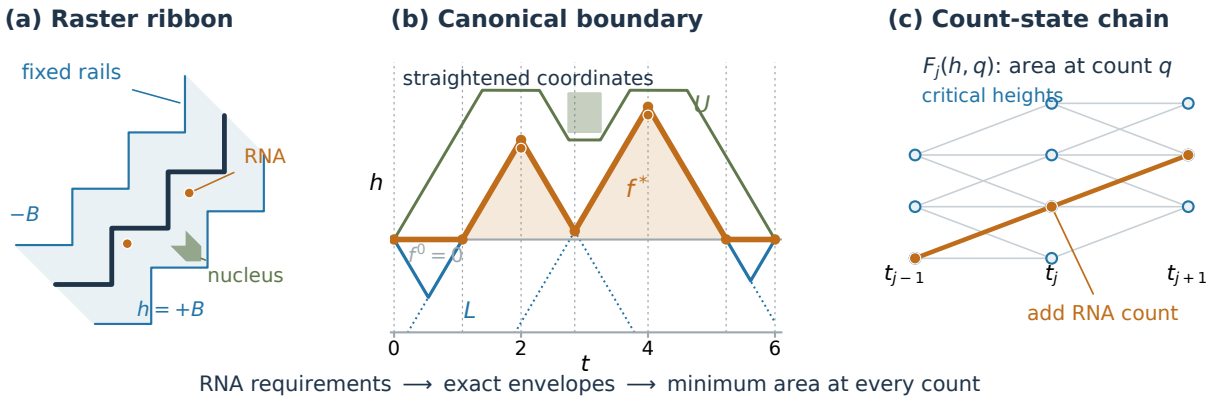



Figure 1: From a shared boundary to an exact count spectrum (schematic). A staircase interface is straightened by a shear chart with fixed outer rails. Molecular requirements propagate as Lipschitz envelopes; clamping the reference between them gives the minimum-area boundary. Their intersections define critical heights for a count-indexed path computation. Theorem 4 converts chart displacement to physical area and glues patch witnesses into one partition.

# 1 Introduction

Imaging-based spatial transcriptomics turns detected molecules into a cell-by-gene matrix through a segmentation. An analyst interpreting a count needs to know whether a plausible change to the segmentation could alter it. Several molecules may move together, use the same physical deformation, or require incompatible boundaries. Independent molecule scores omit some dependencies, and separate local analyses may describe incompatible tissue partitions.

We study a deterministic measurement question: for a fixed reference segmentation and an explicit deformation family, what are the smallest and largest retained counts attainable within an area budget? An answer contains a count interval, its geometric assumptions, and attaining boundaries. A singleton interval certifies invariance within that family. A wider interval quantifies count sensitivity, with a boundary witness explaining each attainable endpoint.

GeoCert resolves two obstacles: preserving the exact area cost of joint molecular assignments when eliminating continuous boundary coordinates, and composing local changes into one tissue partition under one budget. Theorem 3 represents the full assignment-cost function $D^\star(c)$ by a rational cut, supporting general unary objectives and the coupled-interface extension. For a single interface with point transcripts, Theorem 6 uses the same canonical minimum-area boundary to compute every attainable count directly. This count dynamic program recovers states that penalized cuts can miss. Explicit ribbon homeomorphisms make local patch witnesses globally compatible (Figure 1).

**Relationship to existing methods.** Baysor, Proseg, BIDCell and Segger estimate segmentation or transcript assignments [1, 2, 15, 16]. ResolVI models downstream misassignment probabilistically [10]; ovrlpy examines three-dimensional signal overlap [17]. TISSUE constructs uncertainty intervals for predicted, unmeasured expression [19]. Our intervals instead range over geometric alternatives for already observed molecules. They complement these estimators by quantifying sensitivity to a specified family of geometric changes.

Convex segmentation and ordered-label cuts [11, 12], interacting-surface graphs [3, 4], higher-order energy constructions [6, 13], Lipschitz extensions [5], and difference-potential lattices [9] provide foundations. Submodularity alone is insufficient for a binary graph representation with auxiliary variables [14]. We prove the particular cost-preserving elimination below. Exact critical heights offer another realization of that reduction; count augmentation uses established resource-state dynamic programming [20]. Our contribution is the continuous assignment-cost identity and its compatible hard-budget realization.

# 2 Exact elimination of continuous boundary costs

We first establish the continuous assignment-cost identity for one moving interface, the building block of the evaluated atlas. Supplement §1 extends the construction to multiple ordered interfaces with heterogeneous slopes and minimum/maximum thickness constraints.

## 2.1 Requirements and feasible assignments

Let a boundary $f : I \to \mathbb{R}$ on a compact rational interval be $s$-Lipschitz, lie in $[\underline{b}, \overline{b}]$, and have fixed endpoints. The rational piecewise-affine reference $f^0$ obeys these geometric constraints. A site $i$ is a rectangle $[x_i^-, x_i^+] \times I_i$, including a point as a degenerate case. All sites in group $g(i)$ have one owner $c_g \in \{0, 1\}$, with bit $b_g = \mathbf{1}[c_g = 0]$. Clearance $\rho > 0$ requires

$$b_{g(i)} = 1 \Rightarrow f(t) \geq \ell_i = x_i^+ + \rho, \qquad b_{g(i)} = 0 \Rightarrow f(t) \leq u_i = x_i^- - \rho \quad (t \in I_i). \tag{1}$$

Nuclear regions are fixed-label groups. They have the same geometric requirements and zero molecular reward. The reference can satisfy the geometry without satisfying every uncertain site's assignment requirement.

**Lemma 1** (Propagation on one interface)**.** *A lower interval requirement $f \geq \ell_i$ on $I_i$ implies $f(t) \geq \ell_i - s\,\mathrm{dist}(t, I_i)$ everywhere. An upper requirement analogously implies $f(t) \leq u_i + s\,\mathrm{dist}(t, I_i)$.*

*Proof.* Apply the Lipschitz inequality between $t$ and the closest point of the closed interval $I_i$. The displayed bounds are attained by the corresponding tent profiles in the absence of other restrictions. □

Forbid a lower activation if $\ell_i > \overline{b}$ or $\ell_i > f^0(e) + s\,\mathrm{dist}(e, I_i)$ at either endpoint $e$. Forbid an upper activation if $u_i < \underline{b}$ or $f^0(e) > u_i + s\,\mathrm{dist}(e, I_i)$. Add structural pins and

$$b_{g(i)} \Rightarrow b_{g(j)} \quad \text{when} \quad \ell_i - s\,\mathrm{dist}(I_i, I_j) > u_j. \tag{2}$$

This implication excludes simultaneous activation of the incompatible lower and upper requirements.

**Theorem 2** (Exact assignment feasibility)**.** *The finite unary restrictions, implications and structural pins characterize the assignments realizable by the continuous family.*

*Proof.* Necessity follows from Lemma 1. For sufficiency, form the maximum $L$ of the active lower tents, the lower band and both endpoint lower cones. Form the minimum $U$ of the active upper tents, upper band and endpoint upper cones. The pair tests ensure every lower contribution is at most every upper contribution: $\mathrm{dist}(t, I_i) + \mathrm{dist}(t, I_j) \geq \mathrm{dist}(I_i, I_j)$. Unary tests handle bands and endpoints. Thus $L \leq U$. Both envelopes are $s$-Lipschitz, satisfy bands and endpoint equality, and realize the selected requirements. They supply continuous feasible boundaries. □

## 2.2 Preserving every assignment's minimum area

For complete owners $c$, write $\mathcal{F}(c)$ for the realizing boundaries and

$$D^\star(c) = \min_{f \in \mathcal{F}(c)} \int_I |f - f^0|\, dt, \qquad \min \varnothing = +\infty. \tag{3}$$

The later chart supplies its physical-area conversion. Shared movement makes this function nonadditive across transcripts.

**Theorem 3** (Assignment-wise cost representation)**.** *For rational input, one polynomial-size rational graph with assignment vertices $b(c)$, auxiliary vertices $y$, logical hard arcs and a known constant $\kappa$ satisfies*

$$\min_y \mathrm{Cut}_{G_D}(b(c), y) + \kappa = D^\star(c) \tag{4}$$

*for every feasible assignment. Hard-feasible cuts exist exactly for realizable assignments. After adding arbitrary rational unary molecular costs and a rational $\lambda \geq 0$, a minimum cut detects infeasibility or returns optimal owners and an attaining rational piecewise-affine boundary.*

*Proof.* Scalar minimum and maximum preserve the slope, band and endpoint constraints. Consequently

$$f^*(t) = \min\{U(t), \max\{L(t), f^0(t)\}\} \tag{5}$$

is feasible. Every realization lies between $L$ and $U$, so this clamp is pointwise nearest to $f^0$ and minimizes its integrated displacement.

The positive displacement is the maximum of activated functions $[\ell_i - s\,\mathrm{dist}(t, I_i) - f^0(t)]_+$. Negative displacement uses upper requirements and complementary bits. Retain fixed requirements and all uncertain-site requirements, including source-inconsistent ones. At feasible owners, positive and negative displacement cannot both occur at the same point. Split at affine-piece endpoints, source knots, zeros and pair crossings. On a slab, sort the positive heights $h_1 \geq \cdots \geq h_k > 0$, with $h_{k+1} = 0$. If $S_j$ contains the activation labels of the first $j$ heights, then

$$\max_{i \text{ active}} h_i = \sum_{j=1}^{k} (h_j - h_{j+1}) \mathbf{1}[S_j \text{ has an active label}]. \tag{6}$$

Integrate each nonnegative affine difference exactly. Standard OR helpers encode these weighted activation sets; the complementary polarity reverses the construction [6, 13]. Repeated group labels are a single bit. Mandatory terms become forced helpers or the constant $\kappa$. Minimizing auxiliaries therefore gives the integrated clamp for each fixed $b(c)$, proving (4) before molecular costs are specified. Add unary costs by successive differences, shifting negative terms by a recorded constant. Scale rational capacities to integers. Set each numerical hard capacity above the sum of finite capacities *after* the costs and $\lambda$ are chosen. A hard arc in the optimum certifies infeasibility; otherwise (5) reconstructs a realizing boundary. □

**What elimination contributes.** For $M$ constant-piece excursion profiles, relative order changes $O(M^2)$ times. An adjacent swap changes only one prefix, so $O(M^2 + M)$ distinct activation helpers suffice after merging. A common source changes positivity and integral coefficients without changing the profiles' relative order. With $G$ groups and $N$ molecular/structural sites, the scalar graph has $O(G + M^2 + M)$ vertices and $O(G + N^2 + M^2 + M)$ arcs. All intersections and integrated weights have polynomial rational encoding. An event sweep accumulates unchanged layer integrals lazily in $O((M^2 + MJ + J + 1)\log(M + J + 2))$ rational-arithmetic and bitset operations for a $J$-knot reference. This bound concerns integration, separately from operand sizes and optimization. The sweep and coverage gadgets use established machinery [18]; their role here is exact elimination of a continuous minimum-area function.

The coupled extension propagates constraints through boundary-separation paths before forming its envelopes. Their componentwise min/max closure makes the clamp simultaneously feasible, preserving the same identity for every assignment of $m + 1$ labels. Its graph-size and proof details are in Supplement §1. The cut representation supports general molecular objectives and coupled boundaries. For the scalar point-transcript atlas, Section 4 instead augments the canonical clamp's exact critical mesh with count states, solving hard budgets directly without a Lagrangian gap.

# 3 From local interfaces to one compatible tissue model

An *atlas* specifies one compatible uncertainty model for the tissue: disjoint two-label ribbons around reference interfaces, with fixed endpoint traces and outer rails. Each RNA belongs to at most one ribbon; the rest of the partition stays fixed. All cell-count outputs refer to this same family.

Let $P_i$ be closed ribbons with pairwise disjoint interiors and injective charts $\Psi_i : I_i \times [-B_i, B_i] \to P_i$. The reference boundary is transverse coordinate zero, each half lies wholly in its stated reference label, and each RNA belongs to at most one ribbon. An allowed displacement $h_i$ is continuous, zero at both endpoints, and $|h_i| \leq qB_i$ for $q < 1$. Fixed regions have additional label constraints.

**Theorem 4** (Compatible gluing and additive physical area)**.** *Every tuple of admissible ribbon boundaries is induced by one global homeomorphism of the reference partition, identity outside the ribbon interiors. For constant chart Jacobians $J_i > 0$, the physical area whose label changes is exactly $\sum_i J_i \int |h_i|$.*

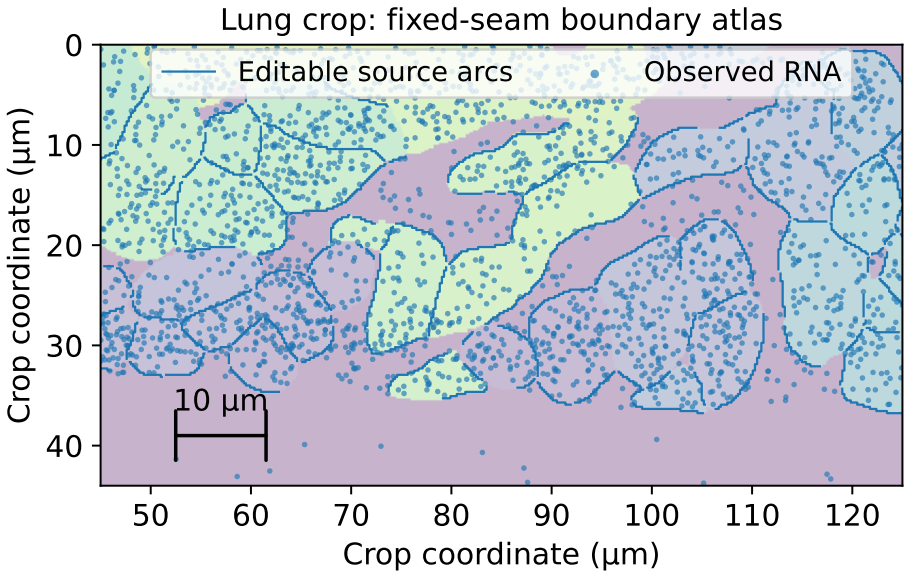



Figure 2: Reference labels and observed RNAs in a lung-crop excerpt. Lines mark editable source arcs; the scale bar is 10 $\mu$m. Ribbons belong to one disjoint atlas. Their endpoint seams, junction neighborhoods and uncovered interfaces stay fixed.

*Proof.* For fixed $t$, define a map of $[-B_i, B_i]$ by

$$H_i(t,v) = \begin{cases} -B_i + (v+B_i)(h_i(t)+B_i)/B_i, & v \le 0, \\ h_i(t) + v(B_i - h_i(t))/B_i, & v \ge 0. \end{cases} \tag{7}$$

The two positive slopes are bounded away from zero; both branches meet at $h_i(t)$ and have continuous explicit inverses. The map fixes rails and is identity at the interval endpoints. Conjugating $(t,v) \mapsto (t, H_i(t,v))$ by $\Psi_i$ gives a homeomorphism fixing the entire ribbon boundary. Extend by identity outside. Finitely many such maps and inverses agree on intersections, so they glue continuously and bijectively. Cell topology and fixed junction incidences are preserved. The changed-label region in one ribbon lies between zero and $h_i(t)$, with chart area $\int |h_i|$. Change of variables and disjoint interiors give the sum. □

**Exact chart for a raster interface.** For a coordinate-monotone raster-edge path choose $s_x, s_y \in \{-1, 1\}$ and let $t = s_x x + s_y y$, $u = s_x x - s_y y$ in pixel coordinates. The exact source arc is $u = r_i(t)$. With pixel size $p$,

$$\Psi_i(t,h) = \frac{p}{2}\big(s_x(t + r_i(t) + h),\ s_y(t - r_i(t) - h)\big), \qquad J_i = p^2/2, \tag{8}$$

up to the recorded origin. This piecewise-affine shear preserves the original staircase interface exactly. Its Jacobian is constant even at changes of source slope, apart from measure-zero corners. The local optimization variable is displacement $h$, with zero source; a source-corner is not a new deformable junction. Intersecting nuclear pixels induce conservative exact interval constraints in this chart. Their prescribed owners remain unchanged.

This family accommodates arbitrary reference adjacency by fixing junction neighborhoods and uncovered perimeter portions. Its topology and coverage are explicit parts of the uncertainty model.

# 4 Hard budgets without assignment enumeration

For a retained count $N_{kg}(c)$, define the minimum-area spectrum

$$S_{kg}(q) = \min_{c:N_{kg}(c)=q} D^\star(c), \quad [L_{kg}(\delta), U_{kg}(\delta)] = [\min, \max]\,\{q : S_{kg}(q) \le \delta\}. \tag{9}$$

The notation means the smallest and largest elements of the displayed feasible-count set. Its interior need not contain every integer. A witness for each endpoint proves attainment; an empty set is reported separately.

## 4.1 Exact critical heights and a count chain

For one boundary, collect all propagated lower/upper requirement pieces, fixed geometric pieces and source pieces before selecting owners. A common mesh contains all piece endpoints and crossings, including source crossings and every point observation. At node $t_j$, retain their candidate values $V_j$ within forced envelopes and discard values violating any point's clearance gap or fixed constraint.

**Lemma 5** (Exact scalar critical mesh)**.** *Restricting nodal heights to these candidates, enforcing original adjacent slope and ownership constraints, and charging trapezoidal absolute displacement preserves $D^\star(c)$ for every feasible assignment.*

*Proof.* Between consecutive events all candidate pieces are affine and their ordering is fixed. The canonical clamp (5) selects one such piece and has constant-sign displacement. Its nodal candidates are retained and its trapezoidal area equals its exact integral. Conversely a feasible candidate-height path interpolates to a feasible continuous boundary: every constraint endpoint is in the mesh, and the intervening constraints are linear. The trapezoid upper-bounds that interpolant's absolute-displacement integral, which is at least $D^\star(c)$. The retained canonical minimizer attains equality. □

**Theorem 6** (Polynomial exact spectra for explicit point RNAs)**.** *Assume one boundary, one point per free RNA group, arbitrary fixed structural intervals, and a nonnegative integer per-point count score. Let $W$ be its variable range and $H_j = |V_j|$. Given the critical mesh and its local node scores, the count dynamic program uses $O((W+1)\sum_j H_j)$ arithmetic transition operations and returns every minimum-area count, with backpointers to attaining boundaries. With explicit unit transcripts the complete algorithm is polynomial in the input and rational encoding.*

*Proof.* A permitted height at a point's mesh node uniquely determines its owner, since the clearance gap was excluded. Count each observation once at that node. Let $z_j(h)$ be its local score and $a_j(h) = w_j|h - f^0(t_j)|$, where $w_j$ is half the sum of adjacent mesh lengths. Define

$$F_j(h,q) = a_j(h) + \min_{u \in V_{j-1}:|u-h| \le s(t_j - t_{j-1})} F_{j-1}(u, q - z_j(h)). \tag{10}$$

Initialization accounts for fixed groups and the first node. The shortest-path induction gives the exact nodal minimum for every count. Lemma 5 gives continuous equality, including unsupported penalty outcomes. For each predecessor count, allowed sorted predecessors form a sliding interval as $h$ increases. A monotone queue computes its minima in $O(H_{j-1} + H_j)$ operations. Sum over counts and nodes. Backpointers reconstruct candidates and hence attaining boundaries. If $P$ is the total affine-piece count, a full mesh has $O(P^2 + P)$ nodes and $O(P)$ candidates per node; this and rational intersection arithmetic give polynomial preparation and encoding. For unit RNA scores, $W$ is bounded by the explicit molecule count. Binary-compressed multiplicities instead give a pseudo-polynomial count dependence. Free interval-valued RNA groups and arbitrary coupled height vectors require additional state and are outside this scalar result. □

Resource-state shortest paths are standard [20]. Here the critical-mesh proof supplies exact continuous costs for each resource state, rather than treating a user-chosen mesh as exact. Single-copy queries use a cheaper corollary: when the source satisfies every source-owner and structural constraint at zero cost, propagate requested bits, keep unforced source bits, and reconstruct the clamp. Every costly requirement in that completion is forced in all feasible completions, so it is minimum-area. The count-chain theorem does not require this stronger source-assignment premise.

### 4.2 One area allowance for the entire atlas

For a fixed cell and gene, compute each incident patch's spectrum $S_i(d)$ in *count changes* relative to its source. Let $n^0_{kg}$ include all retained RNA in that reference cell, including outside the ribbons. Then

$$S_{kg}(q) = \min_{d_1+\cdots+d_r=q-n^0_{kg}} \sum_{i=1}^{r} S_i(d_i). \tag{11}$$

This is min-plus convolution. An attaining trace selects one witness per patch, and Theorem 4 gives a valid global partition. Conversely every allowed partition decomposes into patch states costing at least their local minima. These two directions prove (11). Patches with no relevant gene can stay at their zero-cost source. A simple convolution takes $O(rW^2)$ operations. Thus scalar point-RNA patch atlases give polynomial exact single-cell/gene spectra despite arbitrary fixed reference adjacency.

No molecule or exterior count is duplicated, and $\delta$ is spent once. Every allowed full count matrix lies componentwise inside the table, but different rows' endpoints need not coexist. Joint marker questions require vector spectra or a joint predicate query, rather than multiplying marginal endpoint sets.

### 4.3 Why a penalty path alone is insufficient

An exact cut yielding $m(\mu) = \min_c\{\mu D^\star(c) - N(c)\}$ gives

$$U(\delta) \le \lfloor \mu\delta - m(\mu) \rfloor, \qquad \mu \ge 0. \tag{12}$$

This is weak duality. Feasible witnesses give the opposite bound; a gap is retained unless exact refinement closes it. Applying the same argument to $-N$ bounds the lower endpoint.

A gap occurs even with eight *distinct, explicit unit* transcripts. Take source zero on $[0, 30]$, slope one, band $[-4, 4]$, and clearance $1/10$. At longitudinal centers $5, 15, 25$, place clusters of $2, 3, 3$ RNAs. In a cluster of height $a = 2, 3, 3$, respectively, point $j$ is $(t, x) = (\text{center} + j/100, a - j/100 - 1/10)$. Clearance forces common ownership within each cluster; its first cone dominates, and cluster supports are disjoint. The exact count/area states are $(0, 0), (2, 4), (3, 9), (5, 13), (6, 18), (8, 22)$. At budget ten the maximum is three, whereas the best penalty upper bound is four: for $\mu \le 1/3$, the count-eight state gives $B(\mu) \ge 8 - 12\mu \ge 4$; for $\mu \ge 1/3$, the count-two state gives $B(\mu) \ge 2 + 6\mu \ge 4$. Equality holds at $1/3$. The count chain recovers the unsupported count-three state. The example uses no binary-encoded multiplicity.

## 5 Evaluation: count effects and geometric reach

### 5.1 Observed inputs and fixed extraction rules

Two lung crops contain 7,370 retained Q20 measured-gene records from one public Xenium specimen [7, 8]. Eligibility requires a reference cell to be connected and avoid crop borders. Source labels come from masks. All interfaces between eligible cells, or an eligible cell and background, are considered. The geometry-only rule traces monotone arcs, caps runs at 128 edges, freezes one edge at each end, and retains at least four edges. Longer arcs are processed first with deterministic coordinate ties. The primary atlas tries support halfwidths $B = 4, 2, 1, 1/2$ in chart units, accepting the first feasible width. Exact rational intersection tests certify side labels and disjoint interiors. The displacement band is $|h| \le 3B/4$, slope limit one, and chart clearance $10^{-6}/p$. Nuclear fragments impose fixed-owner interval requirements.

The same rule is applied to two Xenium breast crops containing 81,182 retained RNAs, from the vendor's trimmed two-FOV format-test dataset [21]. This second input tests transfer of the geometric construction and exact computation. Table 1 summarizes both atlases.

### 5.2 Certified invariance and shared-boundary count changes

The exact spectra yield 210,240 interval rows on budgets $0, 0.05, 0.1, 0.25, 0.5, 1, 2, 4\,\mu\text{m}^2$. At budget 4, 2,780/3,304 lung counts (84.1%) and 17,717/22,976 breast counts (77.1%) are certified invariant within the

| Quantity | Lung crops | Breast crops (Xenium) |
|---|---|---|
| Retained RNA | 7,370 | 81,182 |
| Crop-contained connected cells | 153 | 467 |
| Disjoint ribbons | 544 | 1,708 |
| RNA inside ribbons | 745 | 8,262 |
| Editable incident-edge coverage | 68.3% | 65.8% |
| Cell–gene spectra | 3,304 | 22,976 |
| Original eight-budget interval rows | 26,432 | 183,808 |

Table 1: Atlas size and coverage. All eligible reference cells contribute to coverage, including cells with no editable arc. Count spectra include positive or editable cell–gene pairs; RNA outside the ribbons contributes its fixed reference count.

primary atlas. At budget 0.05, these fractions are 88.9% and 84.0%. The remaining intervals quantify count variation and provide attaining boundaries. These totals include fixed counts outside editable ribbons.

A marginal outer bound allows every individually affordable transfer simultaneously. A feasibility-matched additive inner approximation instead charges the sum of singleton transfer areas. The first can overspend the common budget or combine incompatible choices; the second can omit affordable shared movement.

A joint-count difference requires a question with at least two individually movable transcripts. We therefore also report question-level denominators: across the original budget grid, marginal width is at least two in 46 lung and 1,026 breast cell–gene questions. Exact geometry tightens 4/46 (8.7%) and 203/1,026 (19.8%), respectively, affecting 4/153 and 139/467 eligible cells. Across all questions, these fractions are 4/3,304 (0.12%) and 203/22,976 (0.88%). The corresponding tightened *row* counts remain four and 228; additive-inner missed endpoints remain two and 68. On the finer 17-budget grid, the same spectra tighten 12/46 lung questions (26.1%) and 361/1,026 breast questions (35.2%), capturing differences that occur between the original budget samples (Supplement §7).

In breast cell 81, LDHB has retained source count 22. At area $0.25\,\mu\mathrm{m}^2$, the exact interval is $[19, 23]$ versus marginal $[17, 23]$. At area 0.5, it is $[17, 23]$ versus additive-inner $[18, 23]$. An attaining boundary transfers three LDHB molecules through one ribbon and two through another, plus one collateral KRT8 molecule. The physical areas are approximately 0.342327 and $0.141713\,\mu\mathrm{m}^2$, totaling 0.484040. This witness demonstrates both approximation failures within the same geometry (Figure 3).

**Joint marker co-detection.** A joint predicate can ask whether both EPCAM and PTPRC remain detected in one cell under the same deformation. At budget $0.25\,\mu\mathrm{m}^2$, five of seven initially co-detected lung cells and 46 of 48 breast cells retain co-detection throughout the atlas family; the remaining two in each dataset admit a witness that removes it. These joint certificates persist through budget 4 (Supplement §8.4), illustrating how the same geometry answers a multi-gene question.

## 5.3 Budget-dependent variation and saturation

The primary support cap $B \leq 4$ and pixel pitch $p \simeq 0.2125\,\mu\mathrm{m}$ imply a maximum paired-coordinate displacement of

$$\|\Psi(t,h) - \Psi(t,0)\|_2 \leq \frac{3pB}{4\sqrt{2}} \leq 0.4508\,\mu\mathrm{m}. \tag{13}$$

This bound describes a geometric allowance, before nuclei and endpoints restrict it further. It explains why the original family mainly probes submicrometre changes.

From area 0.25 onward, the number of non-singleton intervals remains 524 in lung and 5,259 in breast. Occupancy is different from endpoint saturation: 61 breast intervals widen between 0.25 and 0.5, and three widen between 0.5 and 1. The largest exact budgets required to attain both extreme counts are approximately 0.240525 and 0.573939, respectively. The supplement gives every original budget, total interval width and

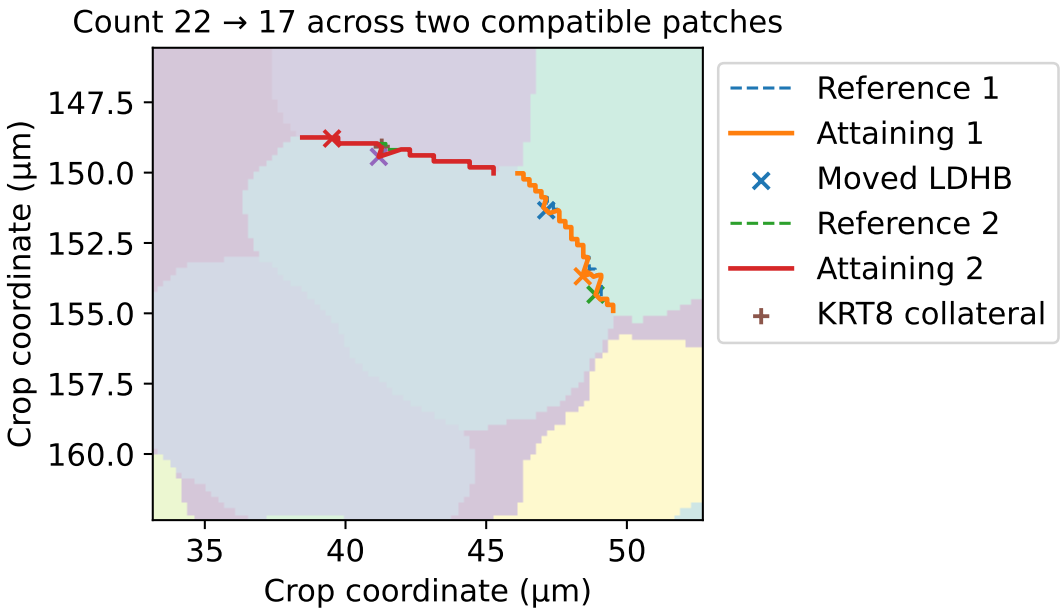



Figure 3: Original reference mask and a count-17 LDHB witness across two compatible patches. Crosses mark five transferred LDHB molecules; the plus marks collateral KRT8. Both changes spend one physical-area budget and coexist in the same partition.

endpoint change count. Thus area controls attainable count changes even after the number of variable intervals stabilizes.

### 5.4 Wider ribbons: reach versus coverage

Before computing width-study outcomes, we fixed physical support halfwidths $r = 0.5, 1, 2, 3\,\mu$m and 17 area budgets from 0 to 16, with finer spacing below 0.5. Each run keeps the original candidate paths, ordering, nuclei, seams and slope. Its required chart halfwidth is $B = \lfloor 4\sqrt{2}r/p \rfloor/4$, with no narrower fallback. Every accepted ribbon therefore obeys the same exact chart and gluing proof. The width-specific atlases are separately selected and need not be nested. This controlled comparison isolates the effect of requiring wider supports.

Table 2 reports the actual displacement cap and incident-edge coverage. At required support 3, the displacement cap reaches about $2.226\,\mu$m, but coverage drops to 16.9% in lung and 17.8% in breast. Local gene multiplicity rises to 17. The last endpoint thresholds now reach 4.878 and $13.783\,\mu\text{m}^2$: hard budgets remain consequential well beyond submicrometre jitter. Wider supports encounter more third-cell interfaces and ribbon conflicts, leaving fewer editable arcs. The primary adaptive extractor retains narrower ribbons where needed; the fixed-width study isolates this coverage cost by disabling fallback.

All five families are evaluated on the same 17-budget grid. The supplement reports each family's interval occupancy, endpoint changes and joint-geometry effects at every budget.

### 5.5 Reach of independent reference assignments

An image-only reference assigns each RNA to the nearest labelled nuclear pixel within $5\,\mu$m, or to background. A five-micrometre crop-edge guard avoids relying on unseen nuclei. We also analyze Proseg 3.2.0 executions [2] on both lung crops. The default factorization failed on the small first crop; one completed stochastic run per crop used factorization disabled, otherwise retaining upstream sampler settings. Proseg assigned 43.0% and

| Support halfwidth | Displacement cap ($\mu$m) | Edge coverage (%) Lung | Breast | Lung Proseg reach (%) | Breast nuclear reach (%) |
|---|---|---|---|---|---|
| Adaptive | 0.451 | 68.3 | 65.8 | 6.6 | 16.2 |
| 0.5 | 0.366 | 45.1 | 44.6 | 4.3 | 11.2 |
| 1 | 0.733 | 33.8 | 34.6 | 5.5 | 14.8 |
| 2 | 1.493 | 23.5 | 24.7 | 4.5 | 14.6 |
| 3 | 2.226 | 16.9 | 17.8 | 3.8 | 12.2 |

Table 2: Required physical support halfwidths are rounded down to quarter chart units. The cap is $3pB/(4\sqrt{2})$; actual feasibility also depends on other constraints. Reach is the fraction of changed reference counts attainable at any budget, with fixed denominators across families.

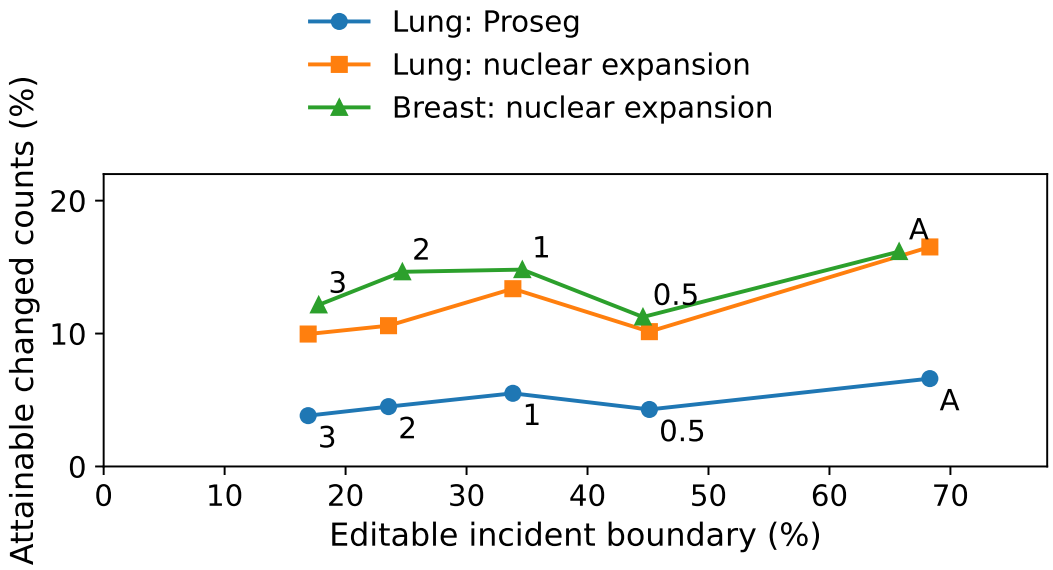



Figure 4: Unbounded-budget count reach versus editable incident-edge coverage on fixed comparison keys. Labels give required support halfwidth in micrometres; A denotes the original adaptive-width atlas. Lines connect separately extracted families, rather than a nested sequence. Proseg is one adapted execution per clipped lung crop.

90.1% of returned molecules to background in the two crops. The comparison tests a broader reassignment regime than local boundary motion. Transcript joins and original-cell mapping were checked against both saved count matrices.

All widths use the same eligible cell–gene comparison keys. Keys involving reference-excluded RNA in any compared atlas are excluded from every width. This leaves 2,798 changed Proseg count entries, and 1,114 lung and 10,965 breast nuclear-reference changes. The primary atlas can attain 185 Proseg changes (6.6%), 184 lung nuclear changes (16.5%) and 1,773 breast nuclear changes (16.2%) at some finite budget. At required support 3, these become 107 (3.8%), 111 (10.0%) and 1,333 (12.2%). Wider local movement therefore loses aggregate reference-count reach under this extraction rule (Figure 4). Countwise inclusion is weaker than simultaneous compatibility, and the complete reference assignment vectors lie outside every tested atlas. A greater area budget cannot repair fixed seams or an unavailable destination cell.

### 5.6 Correctness and operating range

An independent scalar/Cartesian implementation reproduces all 210,240 primary interval rows without calling the production geometry, count chain or optimizer. In the width study, all local witnesses pass continuous checks and all global compositions and interval rows are rechecked. A hash-selected sample of up to 32 multicopy questions per setting (246 total), with two to six target copies, additionally agrees with independent scalar enumeration. The independent verifier uses the same prepared geometry; Proseg execution records are checked separately.

For the unchanged analytical equal-tent scaling family, three fresh processes per size compute every count and check its boundary. Median complete times for 16, 32, 64 and 128 explicit RNAs are 0.465, 1.574, 7.138 and 42.206 seconds. At 128, the method visits 605,632 height/count states and returns all 129 levels without enumerating $2^{128}$ assignments. This controlled scaling experiment demonstrates complete spectrum recovery at sizes beyond exhaustive assignment search. The supplement reports timing environments, resource limits and verification details.

## 6 Discussion and availability

The reduction separates a molecular count question from the least continuous geometry realizing it. Critical heights retain exact minima under count constraints, while compatible ribbons make local witnesses jointly interpretable within a single reference partition. The LDHB example shows why independent transfer permissions and additive transfer costs can err in opposite directions. The width study identifies a different limitation: allowing larger movements within fewer supports can reduce the variety of reference count changes represented. A useful certificate therefore reports its geometric family and coverage alongside its numerical endpoints.

**Limits.** The certificates are exact for the specified family, whose fixed seams preserve topology, nuclear ownership and uncovered boundaries. Anatomical calibration of slopes, widths and budgets is a separate empirical task. The evaluation uses two lung crops from one donor, a breast format-test input intended for computational testing, and one adapted Proseg execution per clipped lung crop; these assess computation and sensitivity rather than assignment accuracy or downstream biological validity. Generalization across donors and whole-slide scaling remain open. Different interval endpoints may require different partitions, so multi-gene conclusions use joint queries. Extending hard-budget count computation to moving junctions and coupled interfaces is a natural next step.

**Software and disclosure.** The MIT-licensed 1.1.0 companion contains the atlas implementation, exact spectra, source crops, Proseg records and reproducible analyses. The source, inputs and results are at `https://github.com/arnavperam/geocert/tree/6581f628a5ac8fb83629f9e0dee0ffa4d685375d/recomb2027-atlas`. The release record identifies the separately rebuilt manuscript and supplement. Source data retain 10x CC BY 4.0 attribution. Generative AI assisted mathematical development, algorithm design, implementation, experiments, verification and writing. The author is responsible for the claims and submission.

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